Week 5 Exercises

Note: The important **prospectus** is due on Oct. 1, so this homework is more brief than other weeks.

Exercise 1 {marginaleffects}

Read Arel-Bundock, Greifer, and Heiss (2024).[1] Write your own cheatsheet for the {marginaleffects} package. *At a minimum*, your cheatsheet should describe commonly used functions their commonly used arguments.

For inspiration: What are the important functions? What are the important arguments? What are good default practices—and how do these deviate from {marginaleffects} defaults? You can stick closely to the perspective in the notes or deviate far from that perspective.

Exercise 2 de Kadt and Grzymala-Busse (2025)

Read de Kadt and Grzymala-Busse (2025), especially Section 3.1, pp. 16-18. There (p. 17), they write:

> If the researcher firmly believes they are not engaged in counterfactual reasoning—they purely want to 'describe the data'—then the use of multivariate regression is itself a peculiar choice.

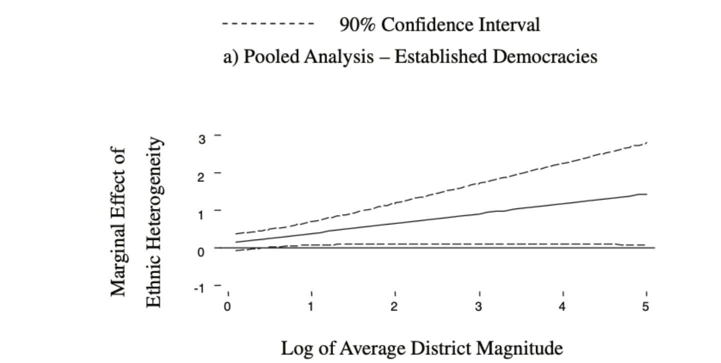
What do you make of their argument? Are control variables always about causal inference? Are control variables ever useful for description? Explain and use examples.

[1] There is also a book at marginaleffects.com/chapters/who.html with lots of examples and case studies.

Exercise 3 Clark and Golder (2006)

Use {marginaleffects} to reproduce *the spirit of*[2] Figure 1 on p. 701 of Clark and Golder (2006), which shows the effect of the effective number of ethnic groups on the effective number of ethnic parties as district magnitude varies.

Figure 1
The Marginal Effect of Ethnic Heterogeneity on the Effective Number of Electoral Parties



The R code below gets you started.

```r
# load data
cg <- crdata::cg2006

# fit model; "Established Democracies 1946-2000" model in Table 2 on p. 698
f <- enep ~ eneg*log(average_magnitude) + eneg*upper_tier + en_pres*proximity
fit <- lm(f, data = cg)
```

Exercise 4 `bioChemists`

Use one of our count models (Poisson, NB, and variants) to defend a useful/interesting/informative *descriptive* claim the number of articles that biochemistry graduate students produced during the last three years. The `bioChemists` dataset in the {pscl} package has the relevant outcome `art`, as well as a number of potential covariates. See `?pscl::bioChemists` for the details.

[2]By "the spirit of," I mean that you should feel free to change the mostly arbitrary features of the plot, like (1) the comparison (the authors use the instantaneous marginal effect), (2) the scale of the x-axis (the authors use the natural log), (3) the values of the other covariates, (4) average case or observed values, etc.

1. With the "peculiar choice" argument of de Kadt and Grzymala-Busse (2025) in mind, select an appropriate set of control variables.
2. Use AIC and/or BIC to select a good model from our menu of options (Poisson/NB; ZI w/ covariates, constant ZI, no ZI; polynomials; interactions).
3. Use {marginaleffects} to compute quantities of interest for all, some, or the best of these models.

```
# load data
biochem <- pscl::bioChemists
```

Exercise 5 Hultman, Kathman, and Shannon (2013)

Hultman, Kathman, and Shannon (2013) use a negative binomial regression model.Skim through their paper to understand the application and the variables. You can find out more about their data and useful details with `?crdata::hks2013`.

1. Does a zero-inflated negative binomial model better fit their data? Consider a model with constant zero inflation and zero-inflation that depends on covariates Z. The Z variables can be the same as the X, or different.
2. Compute a substantively meaningful effect of UN troops on the number of civilians killed. How does the estimate for this quantity of interest change between their negative binomial regression compare to the alternative models you consider?

> ⚠ Warning
>
> The `MASS::glm.nb()` model has a hard time converging for this data. See the `control` options that I use below for the negative binomial model.

```
# load data
hks <- crdata::hks2013

# estimate models
f <- osvAll ~ troopLag + policeLag + militaryobserversLag +
  brv_AllLag + osvAllLagDum + incomp + epduration +
  lntpop

# replicates model 1 in table 1 on p. 884 of HKS
fit <- MASS::glm.nb(f, data = hks,
                    init.theta = 5,
                    control = glm.control(epsilon = 1e-12,
```

```
                                          maxit = 2500,
                                          trace = FALSE))
```

Exercise 6 Radean and Beger (2025)

Read Radean and Beger (2025). The authors make an important point. They argue that reporting only the average effect can be misleading; it can obscure the wide variation in effects across observed values. Instead, they advocate for a case-centered approach, where researchers compute and report effects for each observation.

Use the data and model from Russett and Oneal (2001) in the {crdata} package to illustrate the relative value of a single summary effect (e.g., the average difference across all observations) and the approach recommended by Radean and Beger (2025).

```
# load data
ro <- crdata::ro2001

# glm version of their gee on pp. 314
f <- dispute ~ allies + lcaprat2 + contiguity + dem.lo + logdstab + power
fit <- glm(f, family = "binomial", data = ro)

# example quantity of interest
avg_comparisons(fit,  variables = list(dem.lo = c(-10, 10)))
```

```

 Estimate Std. Error     z Pr(>|z|)     S   2.5 %  97.5 %
  -0.0605    0.00272 -22.3   <0.001 362.3 -0.0658 -0.0552

Term: dem.lo
Type: response
Comparison: 10 - -10
```

Exercise 7 Berry, DeMeritt, and Esarey (2009)

Berry, DeMeritt, and Esarey (2009) have a large dataset and consider a number of polynomials and interactions in their probit model. Using their Model 1 as a starting point, use the AIC and/or BIC to find the best model along two dimensions.

1. **Link Functions**: The authors use `probit`, but other options from within `glm()` are `logit`, `cauchit`, `log`, and `cloglog`.[3] `family = binomial()` uses logit by default; `family = binomial(link = "probit")` uses probit instead. Fit their Model 1 using each of these link functions and use the BIC and/or AIC to find the best. Explain the results.
2. **Interactions and Polynomials**: The authors consider two models. Both have polynomials; one model has interactions as well, the other does not. Consider a simpler model without interactions *or* polynomials. And consider higher-order interactions, deeper interactions, or both. Make sure to use R formula syntax for compact and clear representation. Can you justify a more complicated model that their Model 1 with the IC? Explain the results.

You can find the `scobit.dta` data set here.

TABLE 1 Probit Models of 1984 Presidential Voting Turnout

	Dependent Variable: Pr(*vote*)	
Independent Variable	**(1) Model with Product Terms (i.e., Nagler Specification)**	**(2) Model without Product Terms (i.e., Wolfinger & Rosenstone Specification)**
closing date	0.0006	−0.0078*
	(0.0037)	(0.0004)
education	0.2645*	0.1819*
	(0.0416)	(0.0144)
education²	0.0051	0.0123*
	(0.0042)	(0.0014)
age	0.0697*	0.0697*
	(0.0013)	(0.0013)
age²	−0.0005*	−0.0005*
	(0.0000)	(0.0000)
south	−0.1155*	−0.1159*
	(0.0110)	(0.0110)
gubernatorial election	0.0034	0.0034
	(0.0116)	(0.0116)
closing date ∗ education	−0.0032*	
	(0.0015)	
closing date ∗ education²	0.00028	
	(0.00015)	
constant	−2.7431*	−2.5230*
	(0.1074)	(0.0486)
N	99,676	99,676
Log-likelihood	−55815.28	−55818.03

Standard errors in parentheses.
* $p \le .05$.

```r
# code to load and clean the data
scobit <- haven::read_dta("data/scobit.dta") %>%
  filter(newvote != -1)

# fit author's model 1
f <- newvote ~ poly(neweduc, 2)*closing + poly(age, 2) + south + gov
```

[3]From `?family`: "the binomial family the links `logit`, `probit`, `cauchit`, (corresponding to logistic, normal and Cauchy CDFs respectively) and `log` and `cloglog` (complementary log-log)."

```r
fit1 <- glm(f, family = binomial, data = scobit)

# fit wildly complicated model (51 parameters)
f <- newvote ~ (poly(neweduc, 2) + closing + poly(age, 2) + south + gov)^3
fit2 <- glm(f, family = binomial, data = scobit)

# evaluate w/ bic
BIC(fit1, fit2)
```

```
     df      BIC
fit1 10 111664.2
fit2 51 111707.8
```

References

Arel-Bundock, Vincent, Noah Greifer, and Andrew Heiss. 2024. "How to Interpret Statistical Models Using Marginaleffects for R and Python." *Journal of Statistical Software* 111 (9). https://doi.org/10.18637/jss.v111.i09.

Berry, William D., Jacqueline H. R. DeMeritt, and Justin Esarey. 2009. "Testing for Interaction in Binary Logit and Probit Models: Is a Product Term Essential?" *American Journal of Political Science* 54 (1): 248–66. https://doi.org/10.1111/j.1540-5907.2009.00429.x.

Clark, William Roberts, and Matt Golder. 2006. "Rehabilitating Duverger's Theory." *Comparative Political Studies* 39 (6): 679–708. https://doi.org/10.1177/0010414005278420.

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Hultman, Lisa, Jacob Kathman, and Megan Shannon. 2013. "United Nations Peacekeeping and Civilian Protection in Civil War." *American Journal of Political Science* 57 (4): 875–91. https://doi.org/10.1111/ajps.12036.

Radean, Marius, and Andreas Beger. 2025. "Not-so-Average After All: Individual Vs. Aggregate Effects in Substantive Research." *Journal of Peace Research*. https://repository.essex.ac.uk/40373/.

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